B21-L3 lab: Build a RAG pipeline over survey methodology notes

Platform: Laptop

Work in `labs/rag/` in this repository. It holds five short methodology notes, a gold set of 12 questions, and a package, `survey_rag`, that already handles loading, indexing, prompts and the command line. It needs NumPy and scikit-learn. Seven functions in `src/survey_rag/steps.py` are empty: the ones you wrote in modules B21 and B22. The tests in `tests/` define what each must do. The pipeline runs offline with two keyword rankings. A live model is optional and needs an Anthropic API key. When it works, push it to your own GitHub repository: it is the roadmap's evidence for months 13 to 15.

1. Run `cd labs/rag`, create a virtual environment, then `pip install -e .` and `pip install pytest`. Run `pytest`: most tests fail with NotImplementedError. Read `README.md` and every file in `src/survey_rag/`.
2. Fill in `chunk_words` (lesson B21-L1). Run `pytest tests/test_steps.py -k chunk` until it passes.
3. Fill in `top_k` and `rrf` (lesson B21-L2), then `build_context` and `check_citations` (lesson B21-L3).
4. Fill in `recall_at_k` and `reciprocal_rank` (lesson B22-L1). Run `pytest`: every test passes.
5. Run `python -m survey_rag eval --k 3` and note recall at 3 and MRR. Repeat with `python -m survey_rag --size 40 eval --k 3` and `--size 120`, and keep the size with the higher MRR.
6. Run `python -m survey_rag ask "What changed in the W14 weighting?" --dry-run`. Read the prompt it would send: rules, tagged sources, question.

7. Optional: `pip install -e ".[llm]"`, set `ANTHROPIC_API_KEY`, and run the same command without `--dry-run`. Check the sources and citation problems it prints.
8. Write your README: what the pipeline does, your evaluation numbers, one failure you found and what you changed. Push the folder to a new GitHub repository.

Notes